

# Asif Azad

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## Experience

**AI Innovation Hub, Ministry of Defence Health Services**  
Artificial Intelligence Engineer

Riyadh, KSA | Remote  
April 2025 – Present

- **Co-built and deployed [Digital Harf](#) in a 3-engineer core team**, an accessible multimodal AI platform with 3 adaptive Arabic speech/language therapy modules, using Next.js, TypeScript, Python/FastAPI, Docker, Alibaba Cloud ECS, and GitHub Actions CI/CD.
- **Built an agentic synthetic-data pipeline** with planning, multi-judge quality gates, repair loops, and memory; achieved **90.1% clinical acceptance across 243 judgments by 9 licensed SLPs**.
- **Fine-tuned and benchmarked 3 ASR architectures** for Arabic phoneme prediction; achieved **8.92% PER and 0.004 RTF**, then validated on 40 utterances against 3 SLPs ( $r = 0.791$ ), reducing MAE **16% versus Azure**.
- **Translated clinical requirements into product and evaluation workflows** with SLPs across Middle Eastern centers; co-designed a **90-participant, IRB- and SFDA-approved RCT now underway**.

**Bangladesh University of Engineering and Technology (BUET)**  
Adjunct Lecturer

Dhaka, Bangladesh  
Nov 2025 – May 2026

- **Taught Data Structures & Algorithms, Database Systems, and Digital Logic Design**; designed hands-on labs and mentored students in algorithmic problem solving and software implementation.

## Projects

**Olive AI** | Founding AI Engineer | [oliveai.life](#)

- Architected Arise, an ambient clinical AI pipeline that converts noisy Bangla–English consultations into clinician-reviewable prescriptions and SOAP notes using Gemini, FastAPI, PostgreSQL, and structured outputs.
- Parallelized structured LLM generation and added deterministic medication validation, reducing observed time-to-draft from 60+ seconds to 6–7 seconds (88–90%); engineered streaming, retries, and offline audio recovery for unreliable hospital networks.

**Conformal Evaluation Framework and Platform for Vision–Language Models** | [Findings of EACL 2026](#)

- Built a model-agnostic conformal evaluation pipeline for 18 VLMs across 6 datasets, unifying API/GPU inference and output processing; validated 90% coverage across 3 scoring functions.

## Awards, Honors & Recognition

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|-----------------------------------------------------------------------------------------------------------------------------|------|
| <b>Featured Profile, The Daily Star Campus</b> , recognized for AI-in-healthcare work at Saudi Arabia's Ministry of Defence | 2025 |
| <b>Best Paper Award Nomination</b> , AI in Healthcare Conference (AliH)                                                     | 2025 |
| <b>Silver Award</b> , Blockchain Olympiad Bangladesh 2024   Hyperledger Fabric, Solidity                                    | 2024 |
| <b>IEEE Richard E. Merwin Student Scholarship</b> , sole Fall 2023 recipient from Bangladesh                                | 2023 |
| <b>Champion</b> , Deep Learning Sprint 2023   Computer Vision, Transformer, Finetuning                                      | 2023 |
| <b>Champion</b> , DataVerse Challenge - ITVerse 2023   Natural Language Processing                                          | 2023 |
| <b>Country First</b> , IEEE Xtreme 16.0   Competitive Programming                                                           | 2022 |

## Education

Bangladesh University of Engineering and Technology (BUET)

Feb 2020 – March 2025

B.Sc. in Computer Science & Engineering

CGPA: 3.91/4.00 | Top 10% | Dean's List: 3 consecutive years

Thesis: [Beyond Accuracy: Enhancing Parkinson's Diagnosis with Uncertainty Quantification of Machine Learning Models](#)

## Skills

**Languages:** Python, C++ , Java, TypeScript/JavaScript, SQL, Bash

**AI/ML:** PyTorch, Hugging Face, scikit-learn, NumPy, pandas; model fine-tuning, LLM/VLM evaluation, ASR

**Engineering:** FastAPI, Spring Boot, Next.js, PostgreSQL, Docker, Git, GitHub Actions (CI/CD), Linux

**Google Cloud:** Vertex AI (Gemini API), Compute Engine, Cloud SQL

## Research Interests

My research focuses on building reliable and responsible multimodal intelligent systems that can reason, interact, and operate safely in consequential real-world environments, particularly high-stakes domains such as healthcare.

## Research Expertise

Reliable & Trustworthy AI • Multimodal & Interactive Intelligence • AI for Healthcare & High-Stakes Domains  
• Machine Learning Systems • Generative AI & Agents • Speech & Language AI • Vision–Language Models

## Selected Publications & Research

**PULSAR: Graph-Based Positive Unlabeled Learning with Multi-Stream Adaptive Convolutions for Parkinson's Disease Recognition** Apr 2026

Md Zarif Ul Alam\*, [Asif Azad](#)\*, Md Saiful Islam, Ehsan Hoque, M Saifur Rahman \*Equal contribution

[ACM Transactions on Computing for Healthcare \(Q1, SCImago\)](#) | [Paper](#)

- Co-developed a positive-unlabeled multi-stream graph-convolutional framework for webcam-based Parkinson's screening, modeling noisy labels and joint, bone, velocity, and acceleration streams.

**Validation of Remote Multimodal AI Screening for Parkinson Disease Across Diverse Settings** May 2026

Md Saiful Islam, Tariq Adnan, Abdelrahman Abdelkader, Zipei Liu, Evelyn Ma, Sooyong Park, [Asif Azad](#), et al.

[Communications Medicine \(Q1, SCImago\)](#) | [Paper](#)

- Validated an uncertainty-aware multimodal screening system across **1,865 participants** in supervised and at-home settings, achieving **0.85–0.87 AUROC** across facial, motor, and speech tasks.

**Harf-Speech: A Clinically Aligned Framework for Arabic Phoneme-Level Speech Assessment** Sep 2026

[Asif Azad](#), MD Sadik Hossain Shanto, Mohammad Sadat Hossain, Bdour Alwuqaysi, Sabri Boughorbel, Yahya Bokhari, Abdulrhman Aljouie, Ayah Othman Sindi, Ehsan Hoque

[Interspeech 2026](#) (accepted; CORE A) | [Paper](#)

- Developed and benchmarked a modular Arabic speech-to-phoneme and alignment pipeline, achieving **8.92% PER**, **0.004 RTF**, and strong agreement with speech-language pathologists ( $r = 0.791$ ).

**The Art of Saying “Maybe”: A Conformal Lens for Uncertainty Benchmarking in VLMs** Mar 2026

[Asif Azad](#), Mohammad Sadat Hossain, MD Sadik Hossain Shanto, M Saifur Rahman, Md Rizwan Parvez

[Findings of EACL 2026](#) (EACL: CORE A) | [Paper](#)

- Built a model-agnostic conformal evaluation pipeline spanning **18 open- and closed-source VLMs**, **6 multimodal datasets**, API/GPU inference, standardized output processing, and **3 scoring functions**.

## Selected Publications & Research (continued)

**Digital Harf: A Clinically Integrated Multimodal AI System for Pervasive Arabic Speech and Language Therapy** May 2026  
*Asif Azad, Mohammad Sadat Hossain, MD Sadik Hossain Shanto, Sabri Boughorbel, Abdulrhman Aljouie, Bdour Alwuqaysi, Yahya Bokhari, Ayah Othman Sindi, Ehsan Hoque*  
[Preprint](#) | [Paper](#) | [System](#)

- Co-built a clinically integrated multimodal AI platform and an agentic synthetic-data engine that achieved **90.1% clinician acceptance across 243 judgments by 9 SLPs**; evaluation is underway in a **90-participant RCT**.

**Beyond Accuracy: Enhancing Parkinson’s Diagnosis with Uncertainty Quantification of Machine Learning Models** Aug 2025  
*Asif Azad, Md. Saiful Islam, Ehsan Hoque, M. Saifur Rahman*  
[AIiH 2025, Springer LNCS](#) (Best Paper Award nominee) | [Paper](#) | [Code](#)

- Implemented and compared Monte Carlo Dropout, Deep Evidential Classification, and Bayesian neural networks across motor, facial, and speech datasets for uncertainty-aware Parkinson’s diagnosis.

## Community Service & Leadership

- **Lead Student Coordinator, BUET CSE Festival 2024:** Led multi-month coordination, secured sponsorships, and aligned faculty and student teams to deliver an inter-university Deep Learning Sprint, hackathon, CTF cybersecurity competition, game jam, ICPC-style programming contest, and poster presentation, alongside cultural, indoor/outdoor, and faculty–student programs.
- **Basketball Team Captain, BUET University and Hall Teams (2024):** Captained both teams; led the hall team to the **Inter-Hall championship** as the **tournament’s top scorer**.
- **Reviewer, ACL Rolling Review (2025):** Reviewed research submissions in the October cycle, contributing peer-review service to the computational linguistics community.
- **Invited Speaker & Session Co-Chair (2025):** Presented “AI in Medicine: Transforming Healthcare Today & Tomorrow” and co-chaired the “Power of AI in Medical Science” session at the 52nd Annual National Conference of the Ophthalmological Society of Bangladesh.
- **Workshop Speaker, University of Dhaka (2025):** Conducted “Python for Scientific Research” for the MRS Student Society, introducing students across disciplines to programming and scientific computing.

## Mentoring

- **Mentor & Judge, Solvio AI Hackathon (2025):** Guided 5 of 20 finalist teams through technical and product feedback during a 24-hour final; 3 earned first runner-up, second runner-up, and honorable-mention finishes among 538 competing teams.
- **IEEE Video and Image Processing Cup Mentor (2025):** Mentored 7 BUET CSE undergraduate students on Team BUET Pixelated, which earned a Special Mention in the global competition.
- **Competitive Programming Instructor (2021):** Taught foundational algorithms, data structures, and contest problem solving at BUET’s beginner coding camp.
- **Undergraduate Mentoring, BUET:** Mentored BUET CSE undergraduates during my tenure as an Adjunct Lecturer, holding dedicated one-on-one conversations on career planning and academic choices.

## References

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| <b>Md Saiful Islam, Ph.D.</b><br><i>Thesis Mentor</i><br>Machine Learning Software Engineer<br>Google<br>Mountain View, California<br><a href="mailto:msislam@google.com">msislam@google.com</a> | <b>Prof. M Saifur Rahman</b><br><i>Thesis Supervisor</i><br>Professor<br>Computer Science & Engineering<br>BUET<br>Dhaka, Bangladesh<br><a href="mailto:mrahman@cse.buet.ac.bd">mrahman@cse.buet.ac.bd</a> | <b>Prof. Ehsan Hoque</b><br><i>Research Lead</i><br>Professor<br>Computer Science<br>University of Rochester<br>Rochester, New York<br><a href="mailto:mehoque@cs.rochester.edu">mehoque@cs.rochester.edu</a> |
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